

EL MEHDI HICHAM

Computer Vision Engineer | AI Engineer | Sports Analytics Morocco

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PROFILE

Computer Vision Engineer specialized in **real-time sports analytics** and **multi-object tracking**. Founder and Lead Engineer of **TactiVision**, an end-to-end football intelligence platform leveraging TensorRT, CUDA Streams, homography, pose estimation, OCR, and re-identification to transform broadcast video into tactical insights. Previously contributed to AI systems at SportScore and Priori Sports, with a strong focus on scalable, production-ready computer vision pipelines.

EXPERIENCE

Computer Vision Engineer (Contract)

05/2026 – 06/2026

Priori Sports, Inc. — San Francisco, USA (Remote)

- Developed AI and computer vision solutions for **football spatial tracking**, including homography-based projection systems and tracking pipeline optimization.
- Built **ball trajectory modeling** and airborne ball localization using tracking data and spatial correction techniques.
- Contributed to real-time sports analytics workflows and visualization tools for performance analysis.

Tech: Python, PyTorch, OpenCV, Homography, CUDA, TensorRT, Docker

Founder & Lead Engineer — TactiVision (Independent Product)

2025 – Present

Real-time football analytics platform

- Built and deployed an end-to-end CV pipeline running **7 TensorRT engines in parallel** via CUDA streams on a single RTX 4060: player/GK/referee detection, ball detection, pitch keypoint homography, 17-point pose estimation, field segmentation, jersey OCR, and OSNet re-identification.
- Dedicated goalkeeper model** outputting GK position in real-world metres, speed, skeleton pose and spatial goal coverage on every frame.
- Trained on **140,000+ annotated images** across 7 competitions (LaLiga, Premier League, Ligue 1, Bundesliga, UCL, Botola Pro, FIFA World Cup 2026) — **93.9–98.2% precision**, **~96% accuracy** on unseen footage.
- Live match-centre dashboard generating possession, territory, xThreat, pass networks, heatmaps, pitch control and pressing intensity automatically from **a single broadcast feed** — no stadium infrastructure required.

Tech: TensorRT, ONNX, CUDA Streams, Ultralytics YOLO, ByteTrack, OSNet, OpenCV, PyTorch

R&D Intern — Applied AI & Computer Vision

04/2025 – 10/2025

SportScore, Paris (Remote)

- Built an automatic frame classification module (YOLOv11m-cls) achieving **99.95% accuracy**; implemented complex action detection (pass classification, shots, dribbles, interceptions, goals).
- Optimized the vision pipeline end-to-end: **50% latency reduction** via CUDA Stream parallelization (5 streams) and TensorRT FP16 (7 models), achieving **13 FPS (+117%)**.
- Applied camera homography and geometric transformations for spatial analysis and field mapping.
- Delivered automated performance analytics (heatmaps, pass maps, structured JSON exports, match statistics) and **AI-powered microservices** integrated into production systems.
- Worked autonomously and **mentored 4 interns**, ensuring code quality and consistency.

Tech: Python, PyTorch, TensorFlow, OpenCV, CUDA, TensorRT, Pandas, Docker, FastAPI

Machine Learning Intern — Research & Experimental Evaluation

10/2024 – 01/2025

SETIME Lab, Ibn Tofail University

- Designed reproducible ML pipelines on a **40GB dataset** (2024 Futsal World Cup, 7 matches); conducted controlled experiments with benchmarking and ablation studies.
- Achieved **0.93 recall** using RegNet-Y + Bi-GRU; performed hyperparameter tuning and architecture exploration.

Tech: Python, PyTorch, RegNet-Y, ResNet-50, Bi-GRU, OpenCV, NumPy, Pandas

PROJECTS

OCR System for Handwritten Tifnagh

2024

Hybrid CRNN (CNN + RNN) for Amazigh script recognition; benchmarked against PaddleOCR.

Autonomous Driving — Deep Reinforcement Learning

2024

ROS2 + Gazebo simulation pipeline (camera, LiDAR); TD3 agent for navigation and obstacle avoidance.

SKILLS

Computer Vision: YOLOv8/11, ByteTrack, DeepSORT, OSNet ReID, OCR (PaddleOCR, CRNN), pose estimation, segmentation, homography, camera calibration, 3D geometry

Programming: Python (advanced), C/C++, Shell, SQL, FastAPI, REST API

Deep Learning & ML: PyTorch, TensorFlow/Keras, CNN, RNN/LSTM, Transformers, EfficientNet, RegNet-Y, Bi-GRU

Performance & Deployment: CUDA Streams, TensorRT FP16/INT8, ONNX, real-time inference, Docker, MLOps, AWS (S3, EC2)

Mathematics: linear algebra, geometry, statistical analysis, signal processing, numerical methods

Robotics & Tools: ROS2, Gazebo, LiDAR, camera integration, Pandas, NumPy, Git, Linux/Unix, CI/CD, Agile

Languages: Arabic (native), French (professional), English (professional working proficiency)

EDUCATION

Master's Degree in Artificial Intelligence — Ibn Tofail University, Kenitra

2023 – 2025

Bachelor's Degree in Physics (Theoretical & Mathematical) — Ibn Tofail University, Kenitra

2020 – 2023